Sampling with tfq.layers.Sample() produces same outputs

https://github.com/tensorflow/quantum/issues/550

ROW 124

Sampling with tfq.layers.Sample() produces same outputs #550

New issue

Closed

therooler opened on Apr 28, 2021

Python 3.7
Tensorflow 2.3.1
Tensorflow Quantum 0.4.0

Hi,

I ran into something strange when I was trying to calculate errorbars on the estimators of some observables.
It seems that there are differences between how the `cirq.Simulator()` object handles random numbers compared to how this is done in the default `tfq.layers.Sample()` backend.

Expected behavior
Calling `tfq.layers.Sample()` twice on the same circuit within a script should give me different bitstrings.
Launching the same script multiple times where I sample from a circuit should give me different bitstring for each script.

Actual behaviour
Multiple scripts executed at different times gave me exactly the same set bitstrings. Also, reinitializing the sampling layer within a script can give the same set of bitstrings depending on what backend one uses.

Here is a minimal working example illustrating the problem:

```python
import tensorflow_quantum as tfq
import tensorflow as tf
import cirq
import numpy as np

CASE = 'tfq'

def sample_generator(circuits_tf, repetitions, nqubits):
    # take tfq backend or cirq backend
    if CASE == 'tfq':
        sample_layer = tfq.layers.Sample()
    elif CASE=='cirq':
        sample_layer = tfq.layers.Sample(cirq.Simulator())
    # create generator from sample_layer outputs, repeat `repetition` times.
    data_x = tf.reshape(sample_layer(circuits_tf, repetitions=repetitions).to_tensor(), (-1, nqubits))
    data_x = tf.unstack(data_x)
    for d in data_x:
        yield d


def main():
    # tf.random.set_seed('supcom')

    nqubits = 4
    nsamples = 100
    batch_size = 10
    # create a simple circuit
    qubits = cirq.GridQubit.rect(1, 4)
    circuit = cirq.Circuit()
    circuit.append(cirq.H.on_each(qubits))

    # convert to tfq tensor
    circuits_tf = tfq.convert_to_tensor([circuit])

    # create data generator
    sample_gen = tf.data.Dataset.from_generator(sample_generator, args=(circuits_tf, nsamples, nqubits),
                                                output_types=tf.int32)
    sample_gen = sample_gen.batch(batch_size)
    sample_gen = sample_gen.prefetch(3 * batch_size)

    # get samples and return them
    samples = []
    for batch in sample_gen:
        samples.append(batch.numpy())
    return samples


if __name__ == '__main__':
    samples_1 = main()
    samples_2 = main()
    for s1, s2 in zip(samples_1, samples_2):
        # print(s1, s2)
        print(np.allclose(s1, s2))
```

Assignees
MichaelBroughton

Labels
No labels

Type
No type

Projects
No projects

Milestone
No milestone

Relationships
None yet

Development
No branches or pull requests

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Participants

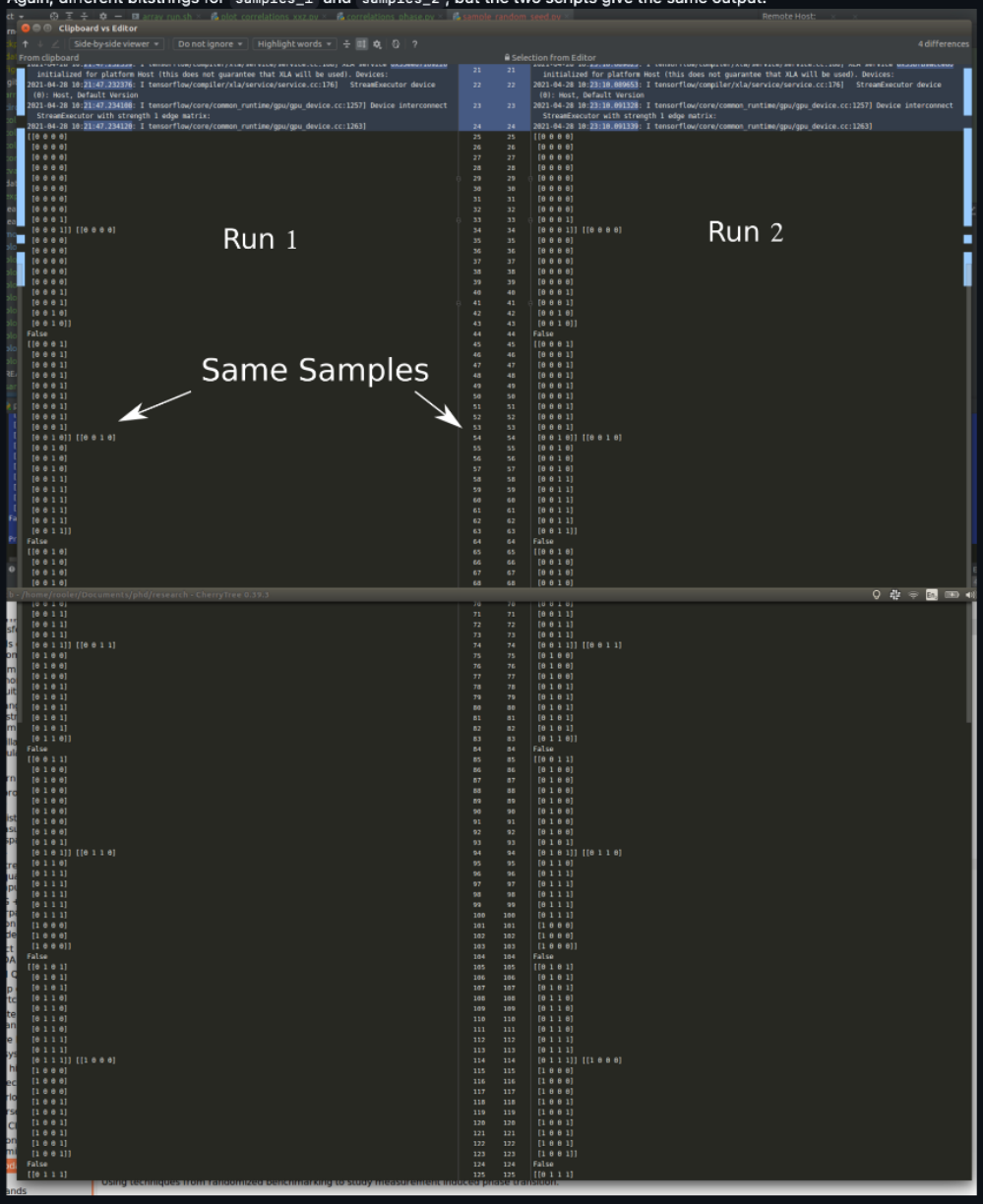
I consider the following cases:

1.) `CASE=='tfq'`

Calling main() twice gives a different set of 100 bitstrings for `samples_1` and `samples_2` (only False is printed in the final line)

2.) `CASE=='tfq'`, two separate script executions.

Again, different bitstrings for `samples_1` and `samples_2`, but the two scripts give the same output:

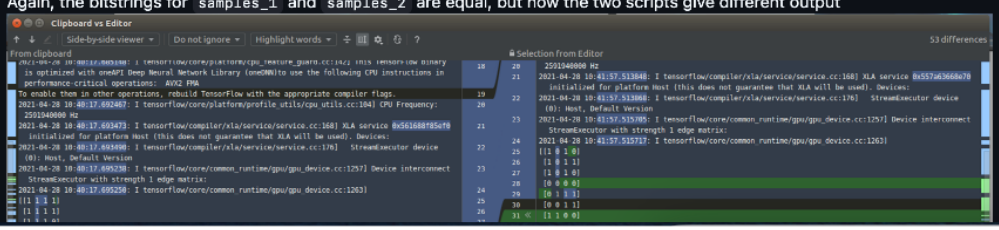


3.) `CASE=='cirq'`

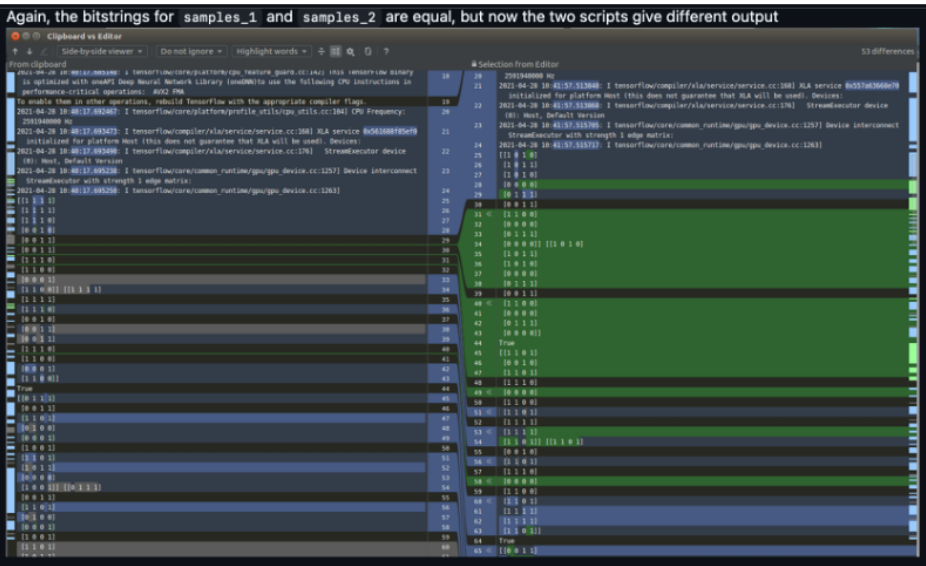
Calling main() twice gives exactly the same 100 bitstrings for `samples_1` and `samples_2` (only True is printed in the final line)

4.) `CASE=='cirq'`

Again, the bitstrings for `samples_1` and `samples_2` are equal, but now the two scripts give different output



Again, the bitstrings for `samples_1` and `samples_2` are equal, but now the two scripts give different output



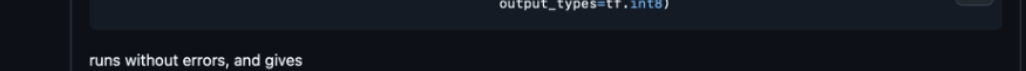
(All of this is unaffected by changing the random seed in Tensorflow between scripts.)

From this I deduce that in TFQ, initializing the Sampling layer initializes a global random number generator once with a fixed seed independent of time (at which the script is executed), and this random number generator is reused for all subsequent calls. This would explain 1.) and 2.).
Whereas with the Cirq backed, the random number generator is reinitialized when the Sampling layer is initialized, but the seed is now dependent on the time (at which the script is executed), which would explain 3.) and 4.).

I fixed my problems by passing a seed to the `cirq.Simulator()` object that depends on time

I this expected behavior with the way I coded this up? Is there a better way to sample bitstrings from the circuit and ensure that everytime I call the sampling layer I get different bitstrings?

Best,
Roeland



MichaelBroughton on Apr 28, 2021 — Collaborator

This might have something to do with #534 . Would you be able to quickly run this same code using `tensorflow==2.4.1` and `tfq-nightly` and see if you are still getting this error ?

therooler on Apr 28, 2021 — Contributor — Author

With `tensorflow==2.4.1` and `tfq-nightly-0.5.0.dev20210428`, changing

```
sample_gen = tf.data.Dataset.from_generator(sample_generator, args=(circuits_tf, nsamples, nqubits),
                                            output_types=tf.int32)
```

to

```
sample_gen = tf.data.Dataset.from_generator(sample_generator, args=(circuits_tf, nsamples, nqubits),
                                            output_types=tf.int8)
```

runs without errors, and gives
1.) Same
2.) Same, but different scripts now give different bitstrings, yay!
3.) Same

1.) Same
2.) Same, but different scripts now give different bitstrings, yay!
3.) Same
4.) Same

MichaelBroughton on Apr 28, 2021 — Collaborator

Hmmmm that still seems like it might be worth investigating. I'm going to look into what we're doing with our randomness generation and see if we need to fix anything up some more.

MichaelBroughton mentioned this on Apr 29, 2021

Switch to using tf philox random. #551

MichaelBroughton self-assigned this on May 3, 2021

MichaelBroughton on May 4, 2021 — Collaborator

@therooler Could you try updating to the latest tf nightly (the one that includes #551) and give an update on the behavior you're seeing ? I'm hoping that this has solved the final issue with the bit strings here.

therooler on May 4, 2021 — Contributor — Author

Using `tfq-nightly-0.5.0.dev20210504` I get the same behavior as in my previous comment.

But this is what we expect right?
1.) Within the same script, calling the sampler twice gives two chains of different samples.
2.) Executing the same script twice gives different samples.

So we never end up with the same bitstrings.

MichaelBroughton on May 5, 2021 — Collaborator

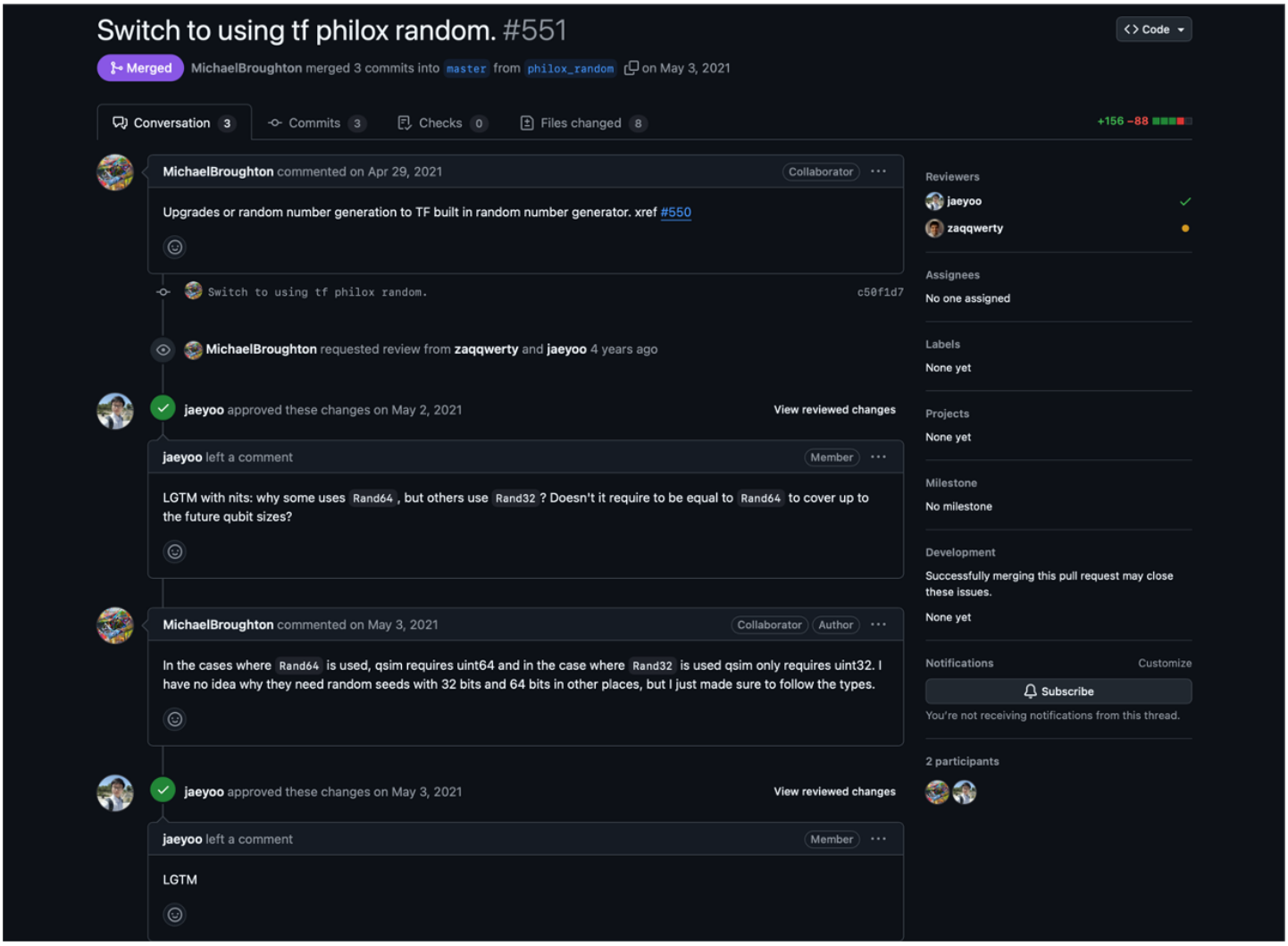
Yes it is what we expect, just wanted to make sure the last changes hadn't re-introcued any bad behavior. I think this is safe to close now.

👍 1

MichaelBroughton closed this as completed on May 5, 2021

jaeyoo added a commit that references this issue on Mar 30, 2023

`Merge pull request tensorflow#550 from quantumlib/release-v0_13_3 ...` Verified `1b3d0f1`


Switch to using tf philox random. #551
Code
Merged
MichaelBroughton merged 3 commits into master from philox_random on May 3, 2021
Conversation 3
Commits 3
Checks 0
Files changed 8
+156 −88
MichaelBroughton commented on Apr 29, 2021
Collaborator
Upgrades or random number generation to TF built in random number generator. xref #550
Switch to using tf philox random.
c50f1d7
MichaelBroughton requested review from zaqqwerty and jaeyoo 4 years ago
jaeyoo approved these changes on May 2, 2021
View reviewed changes
jaeyoo left a comment
Member
LGTM with nits: why some uses Rand64, but others use Rand32? Doesn't it require to be equal to Rand64 to cover up to the future qubit sizes?
MichaelBroughton commented on May 3, 2021
Collaborator
Author
In the cases where Rand64 is used, qsim requires uint64 and in the case where Rand32 is used qsim only requires uint32. I have no idea why they need random seeds with 32 bits and 64 bits in other places, but I just made sure to follow the types.
jaeyoo approved these changes on May 3, 2021
View reviewed changes
jaeyoo left a comment
Member
LGTM
Reviewers
jaeyoo
zaqqwerty
Assignees
No one assigned
Labels
None yet
Projects
None yet
Milestone
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